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Patient-level differential privacy, membership leakage and explanation fidelity in federated brain tumour MRI classification on a deduplicated benchmark

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Highlights

- Deduplicated three-source brain MRI benchmark with patient-disjoint splits
- Record-level DP at ε 8 removes LiRA leakage for 0.08 to 0.13 accuracy
- Patient-level DP needs ε 64 to match record-level ε 8 accuracy
- Unclipped weight noise at matched utility still leaks training patients
- Explanation consistency under DP falls in step with accuracy

Abstract

Background. Federated brain tumour classifiers on public MRI collections report accuracies above 92%, yet the merged sources contain duplicated images, and privacy claims rarely state an accounted budget or a privacy unit.

Purpose. This study measures what formally accounted differential privacy (DP) costs and protects in federated brain tumour MRI classification, at the level of single images and of whole patients, on a deduplicated benchmark.

Methods. Figshare, SARTAJ and Br35H images were deduplicated by perceptual hashing into 3845 images with patient-disjoint splits, and a later correlation check removed 66 mirrored near-duplicates from the test set. A pretrained ResNet-18 was trained centrally and with FedAvg, FedProx and SCAFFOLD under record-level and patient-level DP-SGD, with Rényi accounting at $\delta = 10^{-5}$ and five seeds per setting. Leakage was audited with LiRA using 16 shadow models, and Grad-CAM was scored against expert tumour masks and across seeds.

Results. On 747 clean test images, central balanced accuracy fell from 0.908 without DP to 0.778 at record-level $\varepsilon = 8$, while patient-level DP needed $\varepsilon = 64$ to reach 0.820. Under DP, accuracy declined with client count at fixed data, by 0.096 per unit of $\ln K$ at $\varepsilon = 3$. For federated models without DP, LiRA identified 10.8% of training images at 1% false positives and training patients with AUC 0.805, whereas every DP setting stayed at chance. Unclipped weight noise at matched utility still leaked, and Grad-CAM consistency fell from 0.70 to 0.24 in step with accuracy.

Conclusions. Record-level DP at $\varepsilon = 8$ removes measurable image and patient leakage at a cost of 0.08 (federated) to 0.13 (central) in balanced accuracy, whereas a formal per-patient guarantee costs far more at this data scale.

Keywords Federated learning · Differential privacy · Membership inference · Brain tumour MRI · Explainable artificial intelligence · Data leakage

1 Introduction

Federated learning lets hospitals train a shared tumour classifier without pooling scans, and early brain tumour studies showed that federated models approach the accuracy of centrally trained ones.[1–3] A large body of later work evaluates the same idea on public four-class MRI collections assembled from Figshare, SARTAJ and Br35H, with reported accuracies between 92% and 99%.[4–7]

Those collections overlap. Vangala et al. found that 28.8% of a widely used Kaggle test split has a near-twin in training and that most traceable Figshare test images share a patient with training images.[8] Accuracy measured on such splits partly rewards memorised duplicates, so it cannot show what privacy protection costs a model that must generalise to unseen patients.

Privacy claims in the same literature are seldom checkable. Several federated studies add noise but do not state the privacy budget ε, the failure probability δ or the unit being protected.[9,10] Differentially private stochastic gradient descent (DP-SGD) bounds each example's influence by clipping its gradient before adding calibrated noise.[11] The closest prior work on federated brain tumour MRI instead adds Gaussian noise with standard deviation 0.5 to client weights without clipping, so its stated ε of about 3.4 cannot be derived from the mechanism.[12]

Formal privacy in medical imaging has a measurable price. Kaissis et al. trained a three-site federated model on paediatric chest radiographs at $\varepsilon = 6$, and accuracy fell while remaining comparable to two radiologists.[13] Large budgets can already block reconstruction attacks at small cost.[14] Knolle et al. showed, however, that patients contributing several similar records can face near-perfect membership inference even when an aggregate attack looks random.[15]

That finding matters for slice-based MRI, where one Figshare patient contributes about 13 slices. A guarantee per image then protects a patient only through group privacy, roughly 13 times weaker, so a privacy unit of one patient is the clinically meaningful choice. Explanations raise a second question, since DP is known to degrade gradient-based attributions,[16,17] yet localisation against expert tumour masks under federated DP has not been measured.

This study addresses these gaps on a deduplicated three-source benchmark with expert tumour masks.[18] It asks what record-level and patient-level DP cost in accuracy, how client count and drift correction change that cost, whether measured leakage follows the formal guarantee, and how explanations change under privacy.

1.1 Contributions

1. **A deduplicated, patient-disjoint benchmark.** Perceptual-hash deduplication reduced 10,189 source files to 3845 images split by patient, and an independent correlation check found 66 mirrored near-duplicates in the test set, which are excluded from every reported result.

2. **An accounted privacy-utility map at two privacy units.** Central and federated models were trained under record-level and patient-level DP-SGD with Rényi accounting and five seeds per setting. Record-level $\varepsilon = 8$ cost 0.13 balanced accuracy centrally, while patient-level DP needed $\varepsilon = 64$ for the same utility.

3. **The noise-scale explanation of client count under DP.** Splitting the same data across more clients reduced accuracy under DP, by 0.096 per unit of $\ln K$ at $\varepsilon = 3$, and the aggregated noise scale predicted this decline. FedProx and SCAFFOLD did not recover the loss.

4. **An empirical privacy audit.** LiRA with 16 shadow models per setting detected training images and patients without DP and found nothing above chance under any DP setting. An unclipped weight-noise mechanism at matched utility still leaked.

5. **Explanation fidelity under privacy.** Grad-CAM localisation against expert masks and its consistency across seeds both fell as ε shrank, and the fall tracked accuracy closely ($r = 0.80$).

Section 2 describes the data, training and audit protocol that produced these results.

2 Material and methods

2.1 Data sources and deduplication

Three public collections were combined, namely Figshare (3064 T1-weighted contrast-enhanced slices from 233 patients with expert tumour masks),[18] SARTAJ (3264 images) and Br35H (3861 images). Glioma, meningioma, pituitary tumour and no-tumour labels were kept, which left 7828 images before deduplication.

Each image was rescaled to 8-bit intensity between its 0.5th and 99.5th percentiles, padded to a square and resized to 128×128 pixels with box filtering. A 64-bit discrete cosine transform perceptual hash was

computed per image. Pairs within Hamming distance 4 were treated as duplicates, and pairs within distance 6, together with all slices of one Figshare patient, were linked into one split unit.

Duplicate groups kept one image, preferring Figshare over SARTAJ over Br35H. Groups with conflicting labels were removed (26 images within groups and 23 non-Figshare images conflicting with Figshare), and 48 pairs of Figshare patients sharing a slice were merged into single units. The curated set holds 3845 images in 821 training and test units plus 103 validation units (Table 1).

Units were assigned to training, validation and test splits with seed 2026, giving 2610, 422 and 813 images with no Figshare patient in two splits. As a later check, every pair of curated images was compared by normalised cross-correlation (NCC) of 64×64 versions, both as stored and with one image mirrored.

A test image with NCC of at least 0.95 to any image in another split, a threshold fixed before the check, was marked contaminated. This marked 66 test images, of which 120 of the underlying 148 cross-split pairs matched only after mirroring, a case the perceptual hash cannot detect. All results use the remaining 747 clean test images, and full-set values are given in the supplement.

2.2 Federated partitions

The natural partition (natural3) assigns each source to one client, giving clients of 2029 (Figshare), 351 (SARTAJ) and 230 (Br35H) training images with strong label skew, since no-tumour images come only from SARTAJ and Br35H. Synthetic partitions split the same 2610 images into 10 clients, either uniformly at random or by a Dirichlet prior over class proportions with α of 0.1, 0.5 or 5.0.[19]

To test the effect of client count at fixed data, the $\alpha = 0.5$ partition was also built with K of 3, 5 and 15 clients. For patient-level DP, a Dirichlet partition over whole units ($\alpha = 0.5$, $K = 10$) keeps every patient on one client.

2.3 Model and training

All models are an ImageNet-pretrained ResNet-18[20] whose first convolution sums the RGB filters into one input channel. Batch normalisation layers stay in inference mode with their pretrained statistics, so the network is a per-example function and per-example clipping remains valid, while their affine parameters are trained.

Federated training uses FedAvg[21] with full participation and size-weighted averaging over 100 rounds. Each round, every client runs one local epoch of 10 steps with Poisson sampling at rate $q = 0.1$, stochastic gradient descent with momentum 0.9, a cosine learning-rate schedule and random horizontal flips.

Central training is the one-client case on the pooled training set. The released model is the round with the lowest validation loss. Without DP, gradients are clipped to a global norm of 1.0 for stability.

2.4 Record-level and patient-level differential privacy

Record-level DP-SGD[11] clips each example's gradient g_i to norm $C = 1$, adds Gaussian noise to the sum and divides by the expected batch size qn_j of client j, as in Eq. (1), where B is the sampled batch and σ the noise multiplier.

$$\tilde{g} = \frac{1}{qn_j}\left(\sum_{i\in B} g_i \min\left(1, \frac{C}{\|g_i\|_2}\right) + \mathcal{N}\left(0, \sigma^2 C^2 I\right)\right) \tag{1}$$

The noise multiplier σ was calibrated by bisection so that Rényi DP accounting[22] in Opacus[23] gives the target ε at $\delta = 10^{-5}$ over the 1000 local steps. Targets were ε of 1, 3 and 8, plus a clip-only ablation with $\sigma = 0$ that separates the effect of clipping from that of noise.

Patient-level DP replaces images by units. Units are Poisson-sampled at rate q, the mean gradient of each sampled unit is clipped to C, and noise is added to the sum before division by qU_j, where U_j counts the client's units.

The guarantee then holds per patient, with a unit split across m clients composed over m mechanisms. Central patient-level models used ε of 3, 8, 16, 32 and 64, and federated ones $\varepsilon = 8$. Because learning rates

were chosen on validation data among $k = 2$ candidates trained on the same records, the ε composed over both candidates is also reported, following Papernot and Steinke.[24]

2.5 Drift correction and the weight-noise comparator

FedProx[25] adds a proximal term with weight μ to each local objective, and μ from 0.01 to 10 was screened on validation loss at seed 0. SCAFFOLD[26] uses option II control variates, with two corrections required by this setting.

The normaliser of the control variate uses the displacement of heavy-ball momentum, $\eta\, S(K,\beta)$ with $S = \sum_{k=1}^{K}(1-\beta^k)/(1-\beta)$, instead of $K\eta$, where η is the learning rate, K the number of local steps and β the momentum. The server control variate is averaged with the same size weights as the model, and an ablation keeps the uniform average.

To test the mechanism of the closest prior work,[12] Gaussian noise $\mathcal{N}(0, s^2)$ was added to each client's weights after local training, without clipping or accounting. The literal $s = 0.5$ was kept, and a utility-matched s was chosen by a rule fixed beforehand.

That rule took the largest s in $\{0.5, 0.05, 0.01, 0.003, 0.001\}$ whose seed-0 validation loss did not exceed that of record-level $\varepsilon = 8$ FedAvg on natural3, which selected $s = 0.003$.

2.6 Membership inference audit

The likelihood-ratio attack LiRA[27] was run against natural3 FedAvg at no DP and at ε of 8, 3 and 1, and against the weight-noise comparator. Each target configuration has 16 shadow models trained with the target's own settings and noise multiplier.

Each shadow model trains on a random half of the training and test units, so every patient is wholly in or out. The attack statistic is the logit of the true-class confidence on the image and its mirror, and online LiRA compares Gaussian fits to IN and OUT shadow statistics with one pooled variance per side.

Members are the training images and non-members the clean test images. Patient-level scores sum image scores over a Figshare patient, comparing 126 training with 40 test patients. True-positive rates at 1% and 0.1% false-positive rate are the primary measures, with AUC reported alongside.

2.7 Explanation metrics

Grad-CAM[28] on the last residual block, Grad-CAM++[29] and Integrated Gradients[30] with a black baseline were computed for the 645 Figshare test tumour images that have expert masks. Localisation used the pointing game with a 3-pixel tolerance, the IoU of the top 15% of pixels with the mask and the share of saliency inside the mask.

Faithfulness used deletion and insertion curves over 20 steps.[31] A model-parameter randomisation test[32] checked that maps depend on learned weights, and the outer 8-pixel rim of the brain was scored to detect skull-border artefacts.

Consistency across seeds followed Gupta et al.,[12] with the five seeds of one configuration in place of clients. It is the IoU between the top 10% of each model's Grad-CAM pixels and those of the mean map, computed per image at 8×8 resolution.

2.8 Source controls

Because no-tumour images come only from SARTAJ and Br35H, accuracy was also computed on SARTAJ alone, the only source with all four classes. A logistic-regression probe on penultimate features tested whether the source collection is decodable among images of one class, and the same probe on an ImageNet-initialised and a randomly initialised network gave baselines.

2.9 Statistics and reproducibility

Every setting was trained with seeds 1 to 5, with seed 0 reserved for learning-rate and μ selection. Results are seed means with t-based 95% confidence intervals, and SCAFFOLD was compared with FedAvg by

paired t-tests over seeds with Holm correction across 15 configurations.

The client-count effect was estimated as the slope of balanced accuracy on $\ln K$ over 20 runs per privacy level. Explanation metrics within one model carry 95% intervals from 2000 bootstrap resamples of test patients.

All training ran on Kaggle with two NVIDIA T4 GPUs, about 64 GPU hours in total, from tagged code versions. Each run is identified by a hash of its resolved configuration. The tables and figures were regenerated on Kaggle from the stored run files and frozen with SHA-256 hashes before writing.

These procedures produced the measurements reported next, beginning with the accuracy cost of each privacy setting.

3 Results

3.1 Curated benchmark

Deduplication removed half of all labelled images, mostly because SARTAJ and Br35H repeat each other and Figshare (Table 1). The NCC check then removed 66 test images, almost all Br35H no-tumour images whose mirrored copies sat in training or validation.

Table 1. Images per source through curation and splitting.

Source	Raw files	Four-class labels	After deduplication	Train	Validation	Test	Clean test
Figshare	3064	3064	3010	2029	336	645	644
SARTAJ	3264	3264	505	351	52	102	97
Br35H	3861	1500	330	230	34	66	6
Total	10,189	7828	3845	2610	422	813	747

Removing the contaminated images lowered balanced accuracy by 0.01 to 0.05 in every configuration, and by up to 0.072 in a single run. DP models lost slightly more than non-private ones, so the full test set flattered private models a little more than non-private ones, and all conclusions below hold on both sets (Supplementary Table S3).

3.2 Accuracy cost of record-level and patient-level DP

Record-level DP at $\varepsilon = 8$ reduced central balanced accuracy from 0.908 to 0.778, and accuracy approached chance at $\varepsilon = 1$ (Table 2, Fig. 1). Clipping alone cost 0.063, so most of the loss at $\varepsilon = 8$ came from noise rather than clipping.

Table 2. Balanced accuracy on the clean test set, mean over five seeds with 95% confidence interval.

Setting	No DP	Clip only	$\varepsilon = 8$	$\varepsilon = 3$	$\varepsilon = 1$
Central, record-level	0.908 (0.894, 0.921)	0.845 (0.818, 0.871)	0.778 (0.739, 0.818)	0.648 (0.572, 0.723)	0.326 (0.272, 0.379)
Central, patient-level	0.908 (0.894, 0.921)	0.872 (0.852, 0.891)	0.337 (0.284, 0.390)	0.251 (0.248, 0.253)	not run
FedAvg, natural3	0.841 (0.833, 0.848)	not run	0.765 (0.731, 0.800)	0.674 (0.659, 0.688)	0.525 (0.463, 0.586)
FedAvg, Dirichlet α 0.5, K 10	0.870 (0.852, 0.887)	not run	0.669 (0.626, 0.712)	0.590 (0.548, 0.631)	0.331 (0.280, 0.383)

Patient-level DP was far more expensive at the same ε. Patient-level clip-only training reached 0.872, so weighting each patient once did not itself harm accuracy, but at $\varepsilon = 8$ accuracy fell to 0.337.

The patient-level curve recovered only at large budgets, with 0.565 at $\varepsilon = 16$, 0.727 at $\varepsilon = 32$ and 0.820 at $\varepsilon = 64$. Matching record-level $\varepsilon = 8$ therefore needed a patient-level budget about eight times larger.

The gap follows from the noise scale. Patient-level noise is divided by the number of units, 634 in training, rather than by 2610 images, so the same σ produces about four times more noise in each averaged update.

Record-level $\varepsilon = 8$ is not a weak guarantee per patient by accident. A Figshare patient contributes a median of 13 slices, so group privacy bounds that patient's loss only at roughly 13 times the per-image budget.

3.3 Client count under DP

With the same 2610 images split across more clients, accuracy fell under DP but not without it (Fig. 2). At $\varepsilon = 3$ the slope was -0.096 per unit of $\ln K$ ($p < 0.0001$), from 0.677 with three clients to 0.525 with 15.

At $\varepsilon = 8$ the decline reached -0.156 per unit of $\ln K$ ($p < 0.0001$) with a learning rate of 0.01, but almost vanished with the rate of 0.003 tuned for 10 clients (-0.016, $p = 0.38$). Without DP the slope was $+0.013$ ($p = 0.08$).

With size-weighted averaging, the noise in the averaged update has standard deviation $\sigma C\sqrt{K}/(qN)$ whatever the client sizes, so both privacy levels lie on one curve against this quantity. Client count, not label skew, set the cost of DP here, which also explains why the three-client natural3 partition held up best at $\varepsilon = 1$.

3.4 Drift correction

Neither FedProx nor SCAFFOLD recovered the utility lost to DP. FedProx with μ from 0.01 to 1 changed validation loss by less than one seed standard deviation in all 15 configurations, and $\mu = 10$ made models worse without DP and at $\varepsilon = 8$.

Size-weighted SCAFFOLD did not differ from FedAvg in any configuration after Holm correction, with the smallest adjusted $p = 0.064$. With the uniform server average of the original algorithm, SCAFFOLD collapsed on unequal clients, falling to 0.590 against 0.825 on natural3 at $\varepsilon = 8$ on the full test set.

3.5 Membership inference

Without DP, LiRA identified training images and especially training patients well above chance (Table 3, Fig. 3). At a 1% false-positive rate it found 10.8% of training images, and summed over slices it separated training from test patients with AUC 0.805.

Table 3. Online LiRA against natural3 FedAvg, 16 shadow models, mean over five target models with 95% confidence interval.

Target	Balanced accuracy	Image AUC	TPR at 1% FPR	TPR at 0.1% FPR	Patient AUC
No DP	0.841	0.666 (0.642, 0.689)	0.108 (0.081, 0.135)	0.033 (0.019, 0.046)	0.805 (0.762, 0.848)
$\varepsilon = 8$	0.765	0.464 (0.448, 0.480)	0.017 (0.012, 0.022)	0.006 (0.003, 0.008)	0.454 (0.427, 0.481)
$\varepsilon = 3$	0.674	0.545 (0.522, 0.567)	0.015 (0.008, 0.023)	0.003 (0.002, 0.005)	0.567 (0.529, 0.605)
$\varepsilon = 1$	0.525	0.479 (0.469, 0.489)	0.010 (0.006, 0.015)	0.003 (0.000, 0.005)	0.457 (0.413, 0.500)
Weight noise $s = 0.003$	0.801	0.594 (0.572, 0.616)	0.045 (0.038, 0.051)	0.016 (0.012, 0.020)	0.745 (0.713, 0.778)

Under every DP setting the true-positive rate at 1% false positives stayed between 0.010 and 0.017, the level of a random guess. Average-case AUC under DP moved slightly away from 0.5 in both directions, which reflects the smaller training sets of the shadow models rather than leakage, since the low-false-positive tail stayed at chance.

The weight-noise comparator behaved differently. At $s = 0.5$, the value of the prior work, training diverged in all 21 runs and produced no usable model. At $s = 0.003$ accuracy was 0.801, above FedAvg at

$\varepsilon = 8$, yet the attack found 4.5% of training images at 1% false positives and training patients with AUC 0.745.

A simpler loss-threshold attack was weak even without DP, with AUC 0.52 to 0.54. Computed within each source and class, it stayed below 0.5 under DP, which confirms that its sub-chance values were not caused by the mix of sources in each split.

3.6 Explanations

Grad-CAM localisation against expert masks declined as ε shrank. For correctly classified test images of central models, the pointing-game hit rate fell from 0.197 without DP to 0.156 at $\varepsilon = 8$ and 0.089 at $\varepsilon = 1$, and the top-15% IoU fell from 0.087 to 0.032.

Masks cover 1.8% of the image on average, so these values remain well above chance but far from precise localisation. Grad-CAM placed 0.17 to 0.19 of its mass on the outer brain rim, below the rim's area share of 0.33, and placed less mass outside the brain than the area share of the background.

Consistency across seeds fell from 0.700 without DP to 0.663 with clipping alone and to 0.485, 0.309 and 0.235 at ε of 8, 3 and 1 (Fig. 4a). Patient-level models stayed between 0.22 and 0.29 up to $\varepsilon = 32$ and reached 0.368 at $\varepsilon = 64$.

Across all 115 explained models, consistency correlated with clean balanced accuracy at $r = 0.80$ (Fig. 4b), and top-15% IoU correlated with accuracy at $r = 0.85$. Once accuracy entered a regression, the noise multiplier added no explanatory power for IoU ($p = 0.51$) or consistency ($p = 0.15$).

Integrated Gradients failed the randomisation test, since maps from a fully re-initialised network kept a Spearman correlation of 0.68 to 0.71 with the trained maps. Grad-CAM passed, with correlations between -0.05 and 0.39, so its numbers are the ones interpreted here.

3.7 Source controls

The source collection was linearly decodable from penultimate features in every setting. Among tumour images, a probe separated Figshare from SARTAJ with balanced accuracy 0.913 without DP and 0.851 at $\varepsilon = 1$, close to the 0.871 of the ImageNet initialisation and the 0.846 of a random network.

Source therefore lives in the images themselves, and training added only about 0.04 to it. Accuracy on SARTAJ alone, the one source with all four classes, was 0.663 for central models without DP on the full test set, far below the pooled 0.919, which shows how strongly per-source composition shapes headline accuracy.

These measurements raise questions about why privacy cost so much more per patient than per image, and what the audit does and does not show, which the next section discusses.

4 Discussion

4.1 Protection bought by the privacy budget

The audit and the accounting tell one consistent story. Without DP, a federated model trained on 2610 images leaks enough for LiRA to flag one training image in ten at a 1% false-positive rate, and patient-level aggregation makes the leak clearer still.

Record-level DP at $\varepsilon = 8$ removed every measurable signal, at a cost of 0.08 in balanced accuracy for natural3 FedAvg and 0.13 centrally. This agrees with the view that moderate budgets already defeat practical attacks,[14] although an attack failing does not prove that no stronger attack exists.

The weight-noise comparator separates noise from guarantee. Noise added to weights without clipping can be tuned to keep accuracy, as at $s = 0.003$, but it then leaves training patients identifiable with AUC 0.745. At the published $s = 0.5$ it destroys the model, so neither setting supports a stated ε for that mechanism.[12]

4.2 Higher cost of patient-level protection

Patient-level DP protects the clinically meaningful unit but needs a budget about eight times larger for the same accuracy. The arithmetic is simple, since the noise in each update is divided by 634 units instead of 2610 images, and the clip-only control confirms that noise, not per-patient weighting, causes the loss.

This reinforces the finding of Knolle et al. that multi-record patients are the real privacy risk.[15] At the scale of public brain tumour collections, a strong per-patient guarantee and a useful classifier are not yet compatible, and a larger number of patients is the most direct remedy.

4.3 Client count, not heterogeneity

Under DP, accuracy fell as the same data were split across more clients, while label skew mattered less. The aggregated noise standard deviation $\sigma C\sqrt{K}/(qN)$ grows with the square root of client count, so each added client adds independent noise without adding data.

This mechanism is standard DP-SGD arithmetic and is not claimed as new. Its practical consequence is that cross-silo consortia with few large sites suffer least, and that learning rates tuned for one client count can hide the effect at another.

Drift correction did not help because one local epoch per round produced little drift, while noise dominated the error. The SCAFFOLD ablation also shows that control variates must be averaged with the same weights as the model, or the correction harms exactly the unequal, noisy settings where it is most needed.

4.4 Explanations under privacy

DP degraded Grad-CAM localisation and cross-seed consistency, in line with reports that DP harms gradient-based attributions in general.[16,17] The present results add that, for brain tumour MRI, the degradation follows accuracy closely and the noise level adds nothing once accuracy is known.

For clinical use this means a private model's explanations are about as trustworthy as its predictions, not less. The clip-only control shows that clipping alone barely changes consistency (0.700 to 0.663), so noise drives the change through the accuracy it costs.

The low absolute localisation, with a pointing hit rate near 0.2 even without DP, reflects the 4×4 resolution of the last ResNet block at 128-pixel input. Integrated Gradients with a black baseline failed the randomisation test, which argues against reporting such maps without that check.[32]

4.5 Benchmark leakage and source confounding

The correlation check found mirrored duplicates that perceptual hashing had missed, concentrated in the no-tumour class of Br35H. Training with random flips can then show a model the exact test image, and near-perfect no-tumour specificity on Br35H partly reflected this.

Source is also decodable from images of a single class even by a random network. Headline accuracy on merged public collections therefore partly measures which collection an image came from, as Vangala et al. reported for centralised benchmarks.[8]

Both findings suggest that the 92% to 99% accuracies reported on these collections[4–7] should not be compared with clinical performance. Deduplication by a flip-invariant measure and per-source reporting should become routine for these data.

4.6 Limitations

1. **Remaining duplicates in development data.** Mirrored pairs remain between training and validation (47 pairs), so round selection used slightly contaminated validation data, while all test results exclude contaminated images.

2. **Look-alike units.** SARTAJ and Br35H carry no patient identifiers, so their units are groups of similar images, and the patient-level guarantee is exact only for Figshare.

3. **Shadow-target mismatch.** Shadow models trained on about 1700 images against 2610 for the targets, which shifts average-case AUC, so conclusions rest on low-false-positive rates.
4. **Small patient-level test set.** Only 40 Figshare test patients were available, so patient-level true-positive rates below 2.5% false positives cannot be resolved.
5. **Scope.** All models are 2D ResNet-18 classifiers on axial slices from three public sources, and the federated partitions are simulated rather than real institutions.
6. **Selection cost.** Learning rates were chosen on private validation data, and the ε composed over both candidates is reported but was not used for calibration.

Within these limits, the results support a small number of firm conclusions, stated next.

5 Conclusions

On a deduplicated three-source brain MRI benchmark, record-level DP at $\varepsilon = 8$ removed all leakage that a strong membership attack could measure, at a cost of 0.08 to 0.13 in balanced accuracy. Without DP, the same attack identified training images and, more clearly, training patients.

A formal guarantee per patient cost far more, needing $\varepsilon = 64$ to match record-level $\varepsilon = 8$, because noise is shared among patients rather than slices. Under DP, splitting data across more clients lowered accuracy through the aggregated noise scale, and drift correction did not help.

Weight noise without clipping either destroyed the model or left patients identifiable, so it should not be reported with an ε. Grad-CAM localisation and consistency degraded with privacy in step with accuracy.

Future federated studies on these public collections should use flip-invariant deduplication, state ε, δ and the privacy unit, and test their privacy claims with an attack.

CRediT authorship contribution statement

To be completed by the corresponding author with roles from the 14-role CRediT taxonomy, for example conceptualisation, methodology, software, formal analysis, investigation, data curation, writing of the original draft, review and editing, and visualisation.

Ethics statement

The study used only publicly released, de-identified images from Figshare, SARTAJ and Br35H, and no new patient data were collected. The authors should confirm with their institution that formal ethics approval was not required and state that waiver here.

Declaration of competing interest

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Data availability

Figshare data are available under CC BY 4.0, SARTAJ under the MIT licence and Br35H from its Kaggle page. Split, duplicate and hash manifests, all run results, figure data and code are in the project repository at tag v0.9.1-results-frozen, and the repository visibility and a persistent archive are to be decided by the authors.

Declaration of generative AI and AI-assisted technologies in the writing process

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References

[1] Sheller MJ, Reina GA, Edwards B, Martin J, Bakas S. Multi-institutional deep learning modeling without sharing patient data: a feasibility study on brain tumor segmentation. *Brainlesion*. 2019;11383:92-104. doi:10.1007/978-3-030-11723-8_9

[2] Sheller MJ, Edwards B, Reina GA, et al. Federated learning in medicine: facilitating multi-institutional collaborations without sharing patient data. *Sci Rep*. 2020;10(1):12598. doi:10.1038/s41598-020-69250-1

[3] Pati S, Baid U, Edwards B, et al. Federated learning enables big data for rare cancer boundary detection. *Nat Commun*. 2022;13(1):7346. doi:10.1038/s41467-022-33407-5

[4] Albalawi E, T R M, Thakur A, et al. Integrated approach of federated learning with transfer learning for classification and diagnosis of brain tumor. *BMC Med Imaging*. 2024;24(1):110. doi:10.1186/s12880-024-01261-0

[5] Mastoi QU, Latif S, Brohi S, et al. Explainable AI in medical imaging: an interpretable and collaborative federated learning model for brain tumor classification. *Front Oncol*. 2025;15:1535478. doi:10.3389/fonc.2025.1535478

[6] Muntaqim MDZ, Smrity TA. Federated learning framework for brain tumor detection using MRI images in non-IID data distributions. *J Imaging Inform Med*. 2025;38(6):3909-3927. doi:10.1007/s10278-025-01484-9

[7] Paul M, Tiwari A, Barmashe B, Rai K, Panwar VS. Federated MobileNetV2 with ensemble meta-learning for privacy-preserving brain tumor classification. *Sci Rep*. 2026;16(1). doi:10.1038/s41598-026-55847-5

[8] Vangala BP, Guda S, Peddi L, Vangala N. Scores that hold, benchmarks that leak: measuring dataset contamination in public brain-tumor MRI classification. *arXiv*. Preprint posted online September 30, 2026. arXiv:2610.00421

[9] Lakshmi Vasanthi K, Sree Darshne J, Venkatasubbu P, Ramasubramanian P. A federated multimodal deep learning framework for brain tumor classification using MRI. *Front Artif Intell*. 2026;9:1754000. doi:10.3389/frai.2026.1754000

[10] Qazi EUH, Al-Ghanem WK, Faheem MH, Ullah H. Federated learning framework for privacy-preserving explainable AI-driven clinical decision-making. *IEEE J Biomed Health Inform*. Published online 2026. doi:10.1109/JBHI.2026.3679499

[11] Abadi M, Chu A, Goodfellow I, et al. Deep learning with differential privacy. In: *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*. ACM; 2016:308-318. doi:10.1145/2976749.2978318

[12] Gupta S, Gupta M, Kumar R, Abraham A. A federated learning and explainable AI framework for privacy-preserving brain tumor diagnosis using multi-institutional MRI data. *IEEE Access*. 2026;14:22630-22643. doi:10.1109/ACCESS.2026.3660305

[13] Kaissis G, Ziller A, Passerat-Palmbach J, et al. End-to-end privacy preserving deep learning on multi-institutional medical imaging. *Nat Mach Intell*. 2021;3(6):473-484. doi:10.1038/s42256-021-00337-8

[14] Ziller A, Mueller TT, Stieger S, et al. Reconciling privacy and accuracy in AI for medical imaging. *Nat Mach Intell*. 2024;6(7):764-774. doi:10.1038/s42256-024-00858-y

[15] Knolle MA, Menten MJ, Jungmann F, et al. Disparate privacy risks from medical AI. *Nature*. 2026;656(8126):192-198. doi:10.1038/s41586-026-10688-0

[16] Saifullah S, Mercier D, Lucieri A, Dengel A, Ahmed S. The privacy-explainability trade-off: unraveling the impacts of differential privacy and federated learning on attribution methods. *Front Artif Intell*. 2024;7:1236947. doi:10.3389/frai.2024.1236947

[17] Manna S, Sett N. Reconciling privacy and explainability in high-stakes: a systematic inquiry. *arXiv*. Preprint posted online December 30, 2024. arXiv:2412.20798

[18] Cheng J, Huang W, Cao S, et al. Enhanced performance of brain tumor classification via tumor region augmentation and partition. *PLoS One*. 2015;10(10):e0140381. doi:10.1371/journal.pone.0140381

[19] Hsu TMH, Qi H, Brown M. Measuring the effects of non-identical data distribution for federated visual classification. *arXiv*. Preprint posted online 2019. arXiv:1909.06335

[20] He K, Zhang X, Ren S, Sun J. Deep residual learning for image recognition. In: *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*. IEEE; 2016:770-778. doi:10.1109/CVPR.2016.90

[21] McMahan HB, Moore E, Ramage D, Hampson S, Agüera y Arcas B. Communication-efficient learning of deep networks from decentralized data. In: *Proceedings of the 20th International Conference on Artificial Intelligence and Statistics*. PMLR; 2017;54:1273-1282.

[22] Mironov I. Rényi differential privacy. In: *Proceedings of the 30th IEEE Computer Security Foundations Symposium*. IEEE; 2017:263-275. doi:10.1109/CSF.2017.11

[23] Yousefpour A, Shilov I, Sablayrolles A, et al. Opacus: user-friendly differential privacy library in PyTorch. *arXiv*. Preprint posted online 2021. arXiv:2109.12298

[24] Papernot N, Steinke T. Hyperparameter tuning with Rényi differential privacy. In: *International Conference on Learning Representations*. 2022. arXiv:2110.03620

[25] Li T, Sahu AK, Zaheer M, Sanjabi M, Talwalkar A, Smith V. Federated optimization in heterogeneous networks. In: *Proceedings of Machine Learning and Systems*. 2020;2:429-450.

[26] Karimireddy SP, Kale S, Mohri M, Reddi S, Stich S, Suresh AT. SCAFFOLD: stochastic controlled averaging for federated learning. In: *Proceedings of the 37th International Conference on Machine Learning*. PMLR; 2020;119:5132-5143.

[27] Carlini N, Chien S, Nasr M, Song S, Terzis A, Tramèr F. Membership inference attacks from first principles. In: *2022 IEEE Symposium on Security and Privacy*. IEEE; 2022:1897-1914. doi:10.1109/SP46214.2022.9833649

[28] Selvaraju RR, Cogswell M, Das A, Vedantam R, Parikh D, Batra D. Grad-CAM: visual explanations from deep networks via gradient-based localization. In: *Proceedings of the IEEE International Conference on Computer Vision*. IEEE; 2017:618-626. doi:10.1109/ICCV.2017.74

[29] Chattopadhay A, Sarkar A, Howlader P, Balasubramanian VN. Grad-CAM++: generalized gradient-based visual explanations for deep convolutional networks. In: *2018 IEEE Winter Conference on Applications of Computer Vision*. IEEE; 2018:839-847. doi:10.1109/WACV.2018.00097

[30] Sundararajan M, Taly A, Yan Q. Axiomatic attribution for deep networks. In: *Proceedings of the 34th International Conference on Machine Learning*. PMLR; 2017;70:3319-3328.

[31] Petsiuk V, Das A, Saenko K. RISE: randomized input sampling for explanation of black-box models. In: *Proceedings of the British Machine Vision Conference*. 2018. arXiv:1806.07421

[32] Adebayo J, Gilmer J, Muelly M, Goodfellow I, Hardt M, Kim B. Sanity checks for saliency maps. In: *Advances in Neural Information Processing Systems 31*. 2018. arXiv:1810.03292

Figures

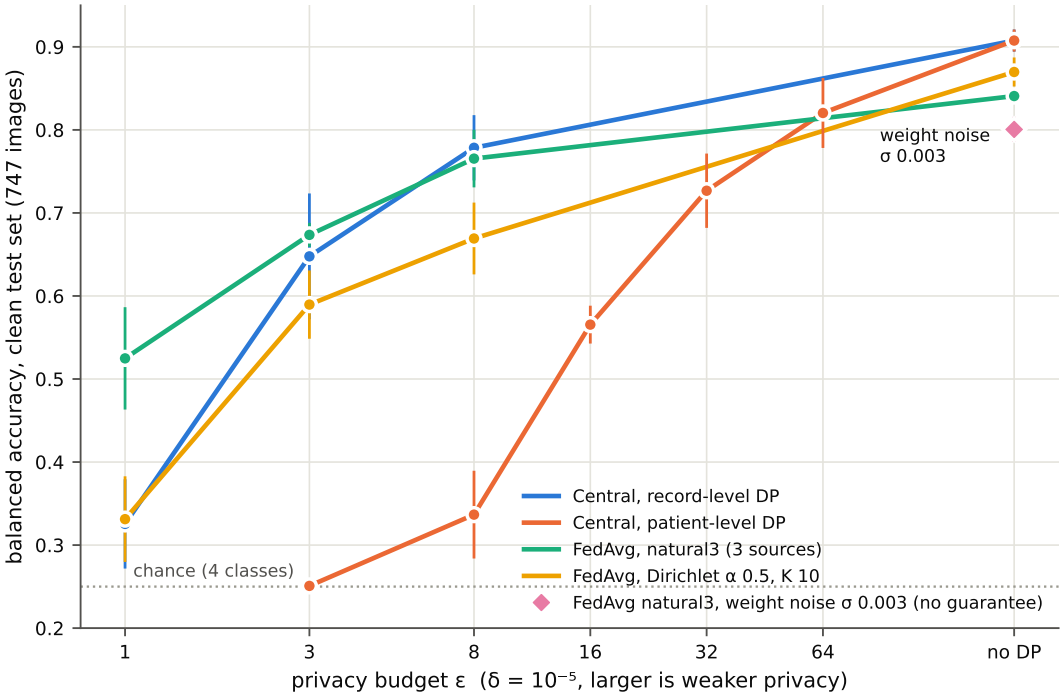


Figure 1. Privacy and utility. Balanced accuracy on the 747 clean test images against the privacy budget ε at $\delta = 10^{-5}$, for central training with record-level and patient-level DP-SGD and for FedAvg on the natural and the Dirichlet $\alpha = 0.5$ partitions. Points are means over five seeds with 95% confidence intervals, and the diamond marks unclipped weight noise ($s = 0.003$), which carries no formal guarantee.

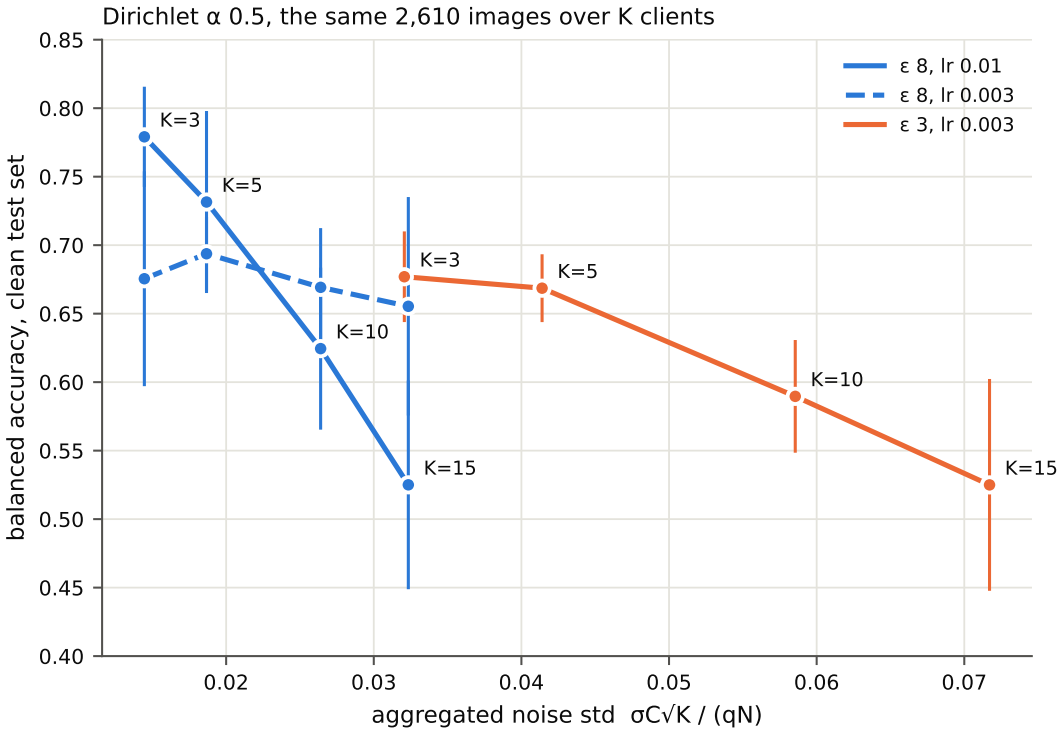


Figure 2. Client count under DP. Clean balanced accuracy of FedAvg on Dirichlet $\alpha = 0.5$ partitions with 3, 5, 10 and 15 clients holding the same 2610 images, against the aggregated noise standard deviation $\sigma C\sqrt{K}/(qN)$. The dashed series uses the learning rate tuned for 10 clients and hides the decline at $\varepsilon = 8$.

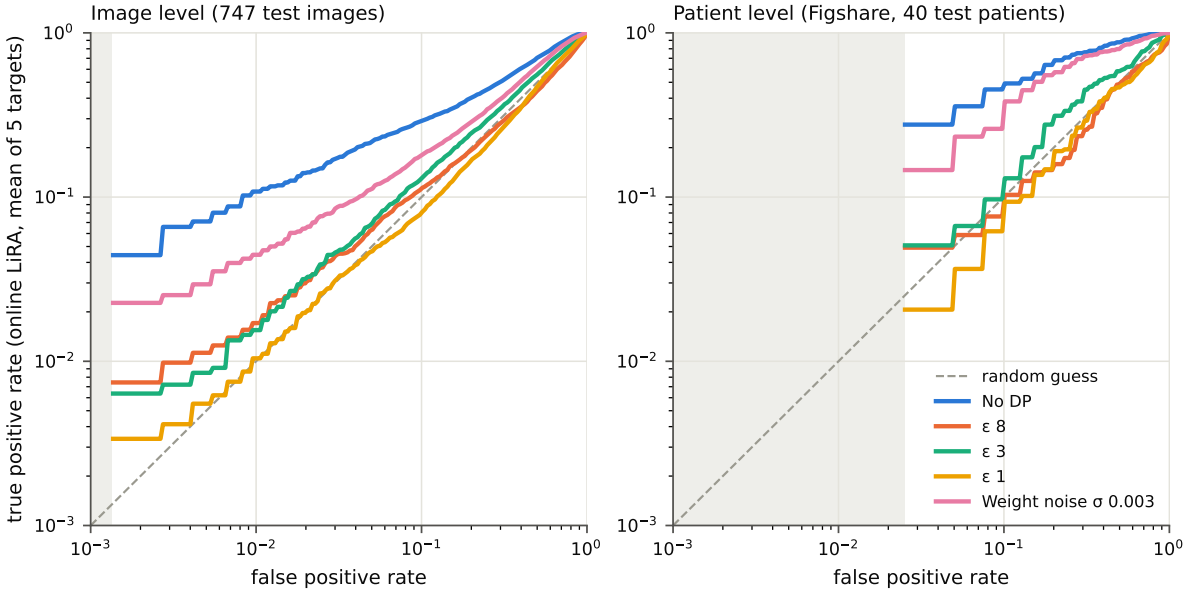


Figure 3. Membership inference. Online LiRA receiver operating characteristic curves on logarithmic axes against natural3 FedAvg, each the mean over five target models with 16 shadow models per configuration. The left panel is image level and the right panel Figshare patient level, and shaded regions lie below the false-positive resolution of the test set.

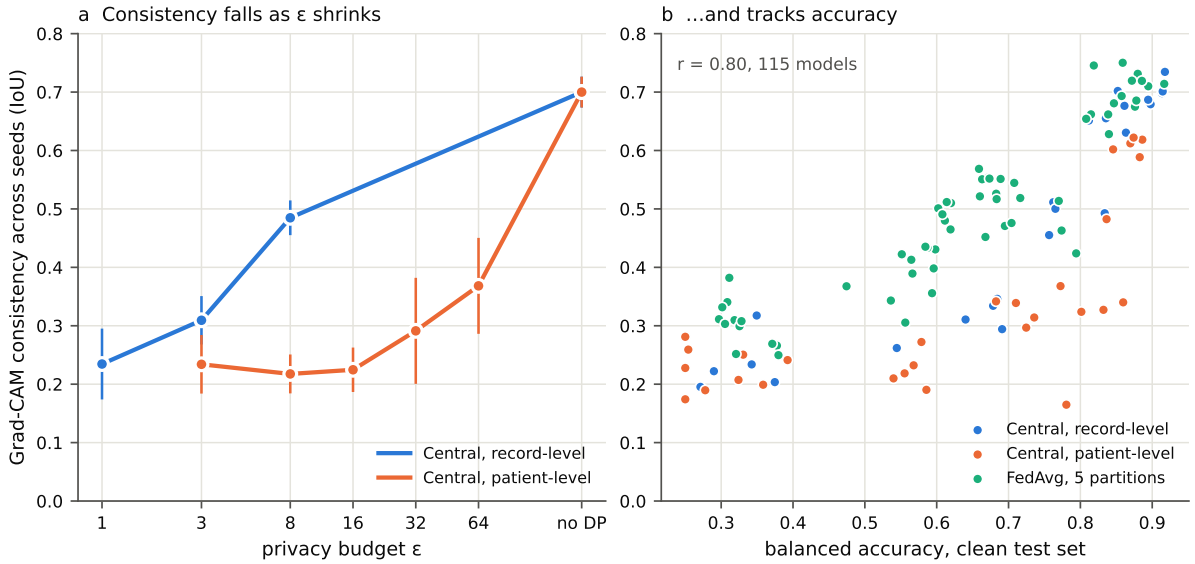


Figure 4. Explanations under privacy. (a) Cross-seed Grad-CAM consistency, the IoU of the top 10% of each model's map with the mean map over five seeds, against ε for central models. (b) Consistency against clean balanced accuracy for all 115 explained models, with Pearson $r = 0.80$.